CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE/IT)
May 2014
CE210 Microprocessor & Computer Architecture

Date: 10.05.2014, Saturday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the following questions. [07]

[A] How many memory locations can be addressed by a microprocessor with 14 address lines? [01]

[B] "Number of machine cycles depends on the size of the instruction." State T/F with justification. [02]

[C] Why processor generates ALE signal at first T state of every machine cycle? [02]

[D] Explain the difference between the machine language and the assembly language of the 8085 microprocessor. [02]

Q - 2 Answer the following questions. [14]

[A] Write down various addressing mode of 8085 microprocessor. [04]

[B] Draw and explain timing diagram of LDA instruction. [04]

[C] Write an assembly language program to perform Logical OR operation without using ORA and ORI Instruction. [04]

[D] Differentiate Memory Mapped I/O Vs Peripheral Mapped I/O. [02]

OR

Q - 2 Answer the following questions. [14]

[A] Calculate the address lines required for an 8K-byte memory chip. And also calculate the number of memory chips needed to design 8K-byte memory if the memory chip size is 1024 X 1. [04]

[B] Draw and explain the timing diagram of OUT instruction. [04]

[C] Write an assembly language program to generate the Fibonacci series of n element. [04]

[D] Differentiate Absolute Decoding Vs Partial Decoding using an example. [02]

Q - 3 Answer the following questions. [14]

[A] Draw and explain functional block diagram of 8085. [05]

[B] Explain following instructions and content of the register(s) after the execution of the instruction: 1. LHLD 2050H. 2. RIM. [05]

[C] What is interrupt? Explain Hardware & Software interrupt. Explain various interrupts of 8085 processor. [04]

OR

Q - 3 Answer the following questions. [14]

[A] 2100 MVI A, 45H [05]

LXI H, 2300H

MOV M, A

What is the size of MOV M, A instruction? How many machine cycle(s) require completing this instruction? Name them. Also Write down the content of AD0-AD7, A0-A7 & A8-A15 for each T-state for MOV M,A instruction only. MOV M, A requires 4 T-sates for opcode fetch cycle.

[B] Explain following instructions and content of the register(s) after the execution of the instruction: 1. SHLD 2050H. 2. XCHG. [05]

[C] What are tri-state devices and why are they essential in bus-oriented system? [04]

SECTION – II

Q - 4 Answer the following questions. **[07]**

[A] How many address lines and data lines are in 8086 microprocessor? [01]

[B] If Accumulator contains C2H and CPI 98H executed, show condition of all flag bits after the execution of this instruction. [02]

[C] Explain why latch is used with an output device and buffer is used with input device? [02]

[D] Which are the pins of 8085 which is related with DMA enabled and an acknowledgement of DMA? [02]

Q - 5 Answer the following questions. **[14]**

[A] Write a short note on Pentium processor. [04]

[B] Describe Mode 0 of 8255 with the help of examples and diagram. [04]

[C] Explain min mode and max mode of 8086 processor. Explain the flag register of 8086. [04]

[D] Write down the features of 8088 microprocessor. [02]

OR

Q - 5 Answer the following questions. **[14]**

[A] Write a short note on 80386 microprocessor. [04]

[B] Draw and explain the block diagram of 8254 programmable interval timer. [04]

[C] Explain transmit & receiver section in UART 8251 IC. [04]

[D] Why stack pointer and program counter is of 16 bit in 8085 microprocessor? [02]

Q - 6 Answer the following questions. **[14]**

[A] Draw and explain block diagram of 8259 programmable interrupt controller with its command word. [07]

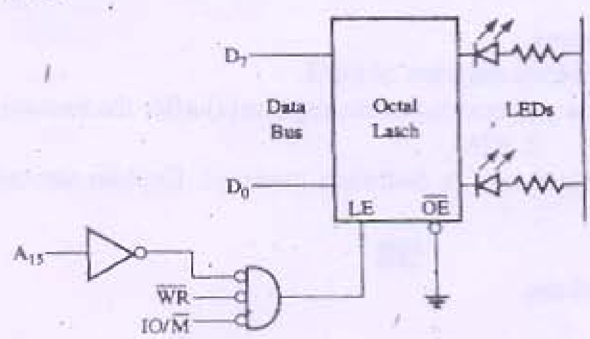
[B] Write a short note on 80486 microprocessor with its block diagram. [07]

OR

Q - 6 Answer the following questions. **[14]**

[A] Draw and explain block diagram of 8279 programmable keyboard/ Display interface in details. [07]

[B] [07]



Identify the port address of the output port in the above figure. Write a program to glow each LED one by one continuously.

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